

Syllabus:- Permutation & Combination**SECTION - I****Multiple Choice Questions (MCQ)**

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

1. How many natural numbers less than a million can be formed using the digits 0, 7 and 8 ?
(1) 728 (2) 729 (3) 485 (4) 486
2. The number of ways of selecting two numbers from the set $\{1, 2, 3, \dots, 30\}$ so that their sum is divisible by 3, is
(1) 95 (2) 145 (3) 190 (4) 435
3. Three straight lines ℓ_1, ℓ_2, ℓ_3 are parallel and lie in the same plane. 5 points are taken on ℓ_1 , 6 points are taken on line ℓ_2 and 7 on line ℓ_3 . What is the maximum number of triangles formed with the vertices at these points ?
(1) 620 (2) 746 (3) 751 (4) 781
4. In a group of 8 boys, two boys are brothers. In how many ways can sit if the brothers are not to sit along with each other ?
(1) $2(6!)$ (2) ${}^7P_2 \times 6!$ (3) ${}^7C_2 \times 6!$ (4) $6!$
5. Consider arrangement of letters AABBBBCDE. Then the number of words in which all B's are together is
(1) 720 (2) 360 (3) 120 (4) 960
6. The number of words that can be formed using all the letter of the word "CURRICULUM" in which two U's are together but separated from the third 'U' is $K(7!)$. The value of K is
(1) 7 (2) 9 (3) 12 (4) 14
7. The number of 3 digit numbers containing digit 7 exactly once is
(1) 225 (2) 280 (3) 270 (4) 243
8. Consider all the 5 digit numbers where each of the digits is chosen from the set $\{1, 2, 3, 4\}$. The number of numbers which contain all the four digits, is
(1) 240 (2) 244 (3) 586 (4) 781
9. A student is to answer 10 out of 13 questions in an examination such that he must chose at least 4 from the first 5 questions. The number of choices available to him is
(1) 140 (2) 196 (3) 280 (4) 346
10. Sum of all the 4 digit numbers formed using digits 1,2,3,4, is (repetition is not allowed)
(1) 33330 (2) 22220 (3) 44440 (4) 66660
11. A basket ball team consists of 12 pairs of twin brothers. On the first day of training, all 24 players stand in a circle in such a way that all pairs of twin brothers are neighbours. Number of ways this can be done is :
(1) $12^{11}(12!)$ (2) $2^{12}(11!)$ (3) $2^{12}(12!)$ (4) $2^{11}(11!)$

12. Find number of three digits number of the form xyz such that $x < y$ and $z \leq y$.
 (1) 156 (2) 204 (3) 240 (4) 276
13. All possible 120 permutations of "WDSMC" are arranged in dictionary order as if each were an ordinary five letter word. The last letter of the 86th word in the list, is
 (1) W (2) D (3) M (4) C
14. If x, y, z are three natural numbers in A.P. such that $x + y + z = 30$ then the possible number of ordered triplet (x, y, z) is
 (1) 18 (2) 19 (3) 20 (4) 21
15. The number of 5 letter words formed using letters of word "CALCULUS" is
 (1) 280 (2) 15 (3) 1110 (4) 56
16. There are six letters $L_1, L_2, L_3, L_4, L_5, L_6$ and their corresponding six envelopes $E_1, E_2, E_3, E_4, E_5, E_6$. Letters having odd value can be put into odd value envelopes and even value letters can be put into even value envelopes, so that no letter go into the right envelopes, then number of arrangement equals.
 (1) 6 (2) 9 (3) 44 (4) 4
17. The total number of ways of selecting six coins out of 20 one rupee coins, 10 fifty paise coins and 7 twenty five paise coins is
 (1) 28 (2) 56 (3) 37/6 (4) None of these
18. Let $A = \{1,2,3,4,5\}$ and $B = \{1,2,3,4,5\}$. If $f : A \rightarrow B$ is an one-one function and $f(x)=x$ holds only for one value of $x \in \{1,2,3,4,5\}$, then the number of such possible function is
 (1) 40 (2) 36 (3) 45 (4) 44
19. 8 players $S_1, S_2, \dots, S_8$, play in tournament. Number of ways in which they can be grouped into 4 pairs so that S_1 and S_2 are in different group is
 (1) 105 (2) 90 (3) 120 (4) 135
20. Let , $N = 2^3 \times 3^5 \times 5^7$, then which of the following statement is false about N?
 (1) Number of proper divisors of N is equal to 192
 (2) Number of divisors of N divisible by 3 but not by 12 equals to 80
 (3) Sum of all divisors of N equals to $(2^4 - 1) \left(\frac{3^6 - 1}{2} \right) \left(\frac{5^8 - 1}{4} \right)$
 (4) If a and b are coprime numbers then number of pairs (a,b) such that $a \times b = N$ are equal to 8

SECTION-II
(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. In a car with seating capacity of exactly five persons, two persons can occupy the front seat and three persons can occupy the back seat. If among seven persons, who wish to travel by this car, only two of them know driving, then the number of ways in which the car can be fully occupied and driven by them, is :
22. Find number of zeros at the end of $^{100}C_{30}$.
23. Find the total number of ways in which 4 balls of different colours can be distributed among 3 persons so that each person gets at least one ball.
24. Find the number of seven digit integers, with sum of digits equal to 10 and formed by using the digits 1, 2 and 3 only.
25. Find the number of ways in which 4 students can sit in 7 chairs in a row if there is no empty chair between any two students.

ANSWER KEY_DPP-24

1.	(1)	2.	(2)	3.	(3)	4.	(2)	5.	(2)	6.	(4)	7.	(1)
8.	(1)	9.	(2)	10.	(4)	11.	(2)	12.	(4)	13.	(2)	14.	(2)
15.	(3)	16.	(4)	17.	(1)	18.	(3)	19.	(2)	20.	(1)	21.	(720)
22.	(1)	23.	(36)	24.	(77)	25.	(96)						